

Hardware · Firmware · Procurement · Integration · Regulatory

Hardware — PCB / Schematic

- ## Hardware — CAD / Mechanical

- ## Firmware

- ✓ `config.h` — all pin assignments and tunable constants
 - ✓ `sensors.h` — TCS34725, ADS1115, SHT31, BH1750, HX711 drivers
 - ✓ `motor.h` — N20 encoder-based motion control + homing
 - ✓ `analysis.h` — HSV-based pad classification (10 analytes)
 - ✓ `printer.h` — CSN-A2 ESC/POS thermal printer output
 - ✓ `main.ino` — state machine: IDLE → HOMING → SCANNING → PRINTING → STERILIZING
 - ■ Flash firmware to hardware and test boot
 - ■ Calibrate HX711 scale factor with known weight
 - ■ Calibrate TCS34725 against Spectralon white standard (update `CAL_R/G/B_WHITE`)
 - ■ Calibrate `TICKS_PER_PAD` with physical dipstick (measure real pad pitch)
 - ■ Validate HSV hue ranges in `analysis.h` using positive/negative reference strips
 - ■ Implement display UI (LVGL or LovyanGFX on DSI LCD)
 - ■ Add Bluetooth/Wi-Fi result upload (optional)
 - ■ Implement OTA firmware update
 - ■ Add watchdog timer and brownout detection
 - ■ Write unit tests for `rgb_to_hsv()` and `classify_pad()`
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BOM / Procurement

- ✓ BOM CSV (`bom/bom.csv`) — 40 line items, estimated total ≈ \$387 per unit
 - ■ Obtain quotes from Indian distributors (Rhydo, Robocraze, Electronicscomp)
 - ■ Source TCS34725 from Adafruit or equivalent (check availability)
 - ■ Procure Labsphere Spectralon calibration standard (long lead time)
 - ■ Verify CSN-A2 printer stock and 58mm paper availability locally
 - ■ Confirm Waveshare 4.3in DSI LCD DSI compatibility with ESP32-S3
 - ■ Bulk pricing target: < \$250 per unit at 50-unit volume
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Integration & Testing

- ■ Power-on test: verify 3V3 and 5V rails under load
 - ■ I2C scan — confirm all 5 devices at expected addresses
 - ■ Motor homing test — both limit switches trigger correctly
 - ■ Full scan cycle with blank dipstick — no crashes, 10 pads detected
 - ■ Full scan cycle with positive control strip — correct analyte results
 - ■ Thermal print output matches on-screen report
 - ■ UV-C sterilization cycle runs for 10 s and shuts off
 - ■ Weight sensor detects cup placement > 1 g threshold
 - ■ Environmental envelope test: 15–35 °C, 20–80% RH
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Documentation

- ✓ Assembly guide template (hardware/schematic/ + GUIDE.md in source data)
 - ■ Write complete assembly guide with photos
 - ■ Write calibration SOP (white standard → pad validation)
 - ■ Write operator quick-start card (print on 58mm paper)
 - ■ Add troubleshooting section (I2C errors, motor faults, print jams)
 - ■ Create block diagram image for README
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Regulatory / Quality (future)

- ■ CE / BIS marking assessment
- ■ IEC 61010-1 electrical safety review
- ■ ISO 13485 QMS alignment checklist
- ■ 510(k) pathway assessment (if exporting to US)